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Real-Time and Continuous Shale Shaker Monitoring Using Machine Learning, Computer Vision and Edge Computing

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Abstract

This paper introduces a machine learning-based system for continuous, real-time monitoring of shale shakers to support early detection of wellbore instability. By combining high-resolution camera feeds with edge computing, the system enables on-site image processing, significantly reducing bandwidth requirements and providing timely insights into shaker conditions. A Vision Transformer (ViT) architecture is used to classify images into three categories—Empty, Fluid, and Cuttings. Image preprocessing techniques, including masking and motion detection, enhance model performance by isolating relevant activity. The model was trained and validated on expert-labeled data from two wells and deployed on edge devices for real-time execution. It achieved 100% classification accuracy, with model predictions strongly correlating with rig activities, including detection of operational anomalies. This automated approach eliminates the limitations of manual inspections, offering a scalable and efficient tool to reduce non-productive time, improve safety, and optimize hole cleaning.

Introduction

Optimizing drilling efficiency while ensuring operational safety and minimizing costs remains a central objective in the oil and gas industry. A significant portion of Non-Productive Time (NPT), periods when drilling either halts or progresses very slowly, comes from wellbore instability, which alone accounts for roughly 10 percent of all NPT events and can cost operators millions of dollars per well (Emhanna 2018; Li et al. 2024). Industry estimates suggest that annual economic losses due to borehole instability exceed USD 1 billion, with lost time representing over 40 percent of drilling-related NPT (Li et al. 2024; Aadnøy and Looyeh 2011). If not detected and mitigated on time, instability can escalate to stuck pipe, equipment damage, or even loss of the well, highlighting the need for more proactive and reliable monitoring strategies.

Shale shakers, as the primary solids control units on a drilling rig, offer an indirect yet timely window into downhole conditions (Marshall 1978). By separating coarse cuttings from the circulating drilling fluid, shakers maintain optimal mud properties and help prevent cuttings buildup. Changes in the volume, size, or appearance of discharged cuttings and the condition of the shaker screens themselves, serve as early indicators of lithological changes or emerging wellbore issues. However, traditional methods of monitoring shale shakers rely on periodic visual inspections by rig personnel, leading to delayed or inconsistent detection of critical events such as screen blinding or cuttings overloads (Ruggiero et al. 2024).

To overcome these limitations, this paper presents a novel, automated monitoring system that leverages machine learning and edge computing for continuous, real-time analysis of shale shaker performance. By deploying a Vision Transformer (ViT) model on an on-site edge device, high-resolution camera feeds are processed locally to classify shaker-screen conditions and flag anomalies without needing high-bandwidth connections to remote servers. This approach delivers objective, uninterrupted data streams, enabling earlier detection of deviations from optimal operating conditions and reducing reliance on intermittent human observations. The primary objective is to detail the design, implementation, and validation of this system, and to demonstrate how integrating ML-driven decision support can enhance wellbore stability, reduce NPT, and optimize hole cleaning during drilling operations.

Background

Wellbore Instability

Wellbore instability arises when stresses around the drilled borehole exceed the formation's strength or when unfavourable interactions occur between drilling fluid and rock (McLean 1990). Mechanical factors, such as high in-situ stresses (often worsened by tectonic activity or geopressured zones), inadequate mud weight leading to collapse, or excessive mud weight inducing fractures, can all trigger instability. The consequences range from ineffective hole cleaning, excessive caving, and stuck pipe to the inability to deploy casing or logging tools, and in extreme scenarios, loss of the well section or entire wellbore. These failures consistently generate significant NPT and inflate operational costs. Early warning signs, such as over-gauge or under-gauge hole measurements, splintery cuttings at the surface, gradual increases in torque and drag, or unexpectedly high circulating pressures, are often subtle and can be masked by routine drilling anomalies, making sensitive, continuous monitoring essential (McLean 1990).

Shale Shaker Operations and Indicators

Shale shakers serve as the first line of defense in solids control by mechanically separating larger drilled cuttings from the returning drilling fluid (Bailey et al. 2024; Plumb et al. 2000). Their efficiency directly influences drilling fluid properties, downhole tool wear, and overall drilling activities (van Oort 2003). The characteristics of cuttings discharged volume, particle size, shape, and angularity provide diagnostic clues about bit performance, lithology changes, and hole-cleaning effectiveness (Plumb et al. 2000). For example, a surge in fine cuttings can signal bit wear or inefficient drilling, while a sudden influx of large, angular cuttings (cavings) often indicates emerging wellbore instability (Zoback 2007).

To achieve best results, the shale shaker activities must be evaluated in direct correlation with rig operations. During active drilling, cuttings should appear on the shaker in a steady, continuous stream. On the other hand, when drilling pauses, such as during connections or tripping, the fluid circulation stops and the shaker should ideally show neither fluid nor solids. Any unexpected presence of cuttings or fluid during these non-circulation intervals, or an abrupt reduction in cuttings flow while drilling is ongoing, can indicate an early warning of drilling anomalies and emerging wellbore issues.

Problem Statement and Existing Monitoring Challenges

The timely detection of changes in shale shaker output is critical for proactive drilling management. However, current monitoring practices face significant hurdles, limiting their effectiveness.

Limitations of Manual Monitoring

Conventional shale shaker monitoring is conducted manually by mud loggers or drilling crew members through periodic visual inspections and cuttings-sample collection (Ruggiero et al. 2024). These intermittent checks are inherently limited: critical events occurring between inspections can be missed or detected too late, and assessments depend heavily on individual experience, leading to variability in interpretation. Moreover, personnel must frequently work in hazardous areas, exposed to high noise, vibration, and drilling fluid aerosols, raising safety concerns. This method is further flawed because the individual responsible for shaker checks may be called away to assist with a connection, become distracted during key changes in drilling parameters (such as flow adjustments), or otherwise be unable to maintain continuous focus on shaker performance.

Challenges with Previous Automated/Sensor-Based Approaches

To address the shortcomings of manual methods, various sensor and camera-based solutions have been trialed. Vibration sensors, flow meters, and acoustics have shown promise, but early demonstrations highlighted the advantages of vision-based approaches. Camera systems have flagged cuttings overloads and shale cavings before traditional surface sensors reacted, though most implementations relied on convolutional neural networks (CNNs) running on remote servers (Bell 2025). For instance, pilot projects in Norway streamed HD video to cloud-based CNNs for inference, while other case studies mounted GPUs in data vans but still needed 2–5 Mbps uplinks per shaker lane bandwidth allocations that proved impractical on many rigs. More recent 3-D scanning systems have improved spatial resolution yet remained tethered to external servers for reconstruction and classification, which limits real-time utility when VSAT bandwidth diminishes or is allocated elsewhere (Prez et al. 2023). These bandwidth and latency constraints have motivated a shift toward compact, on-site inference engines capable of executing deep-learning models locally and transmitting only low-rate metadata.

Enabling Technologies for Advanced Monitoring

Recent advancements in artificial intelligence and computing hardware offer new possibilities for overcoming the limitations of traditional and early automated monitoring systems. This project leverages several key technologies.

Machine Learning and Computer Vision in Drilling

Machine learning (ML) techniques are increasingly employed throughout the oil and gas sector to analyze complex datasets and enhance decision-making. In drilling, ML has been applied to optimize rate of penetration (ROP), predict stuck-pipe incidents, and perform real-time drilling parameter optimization, formation-top identification, and wellbore stability assessment from drilling and logging data. Within computer vision, a subset of ML, applications include automatic detection of bit cutter damage and image-based analysis of cuttings on shale shakers to classify particle types, sizes, shapes, and detect problematic conditions like cavings (van Oort 2003; Zoback 2007). Vision-based solutions promise faster, more consistent, and safer alternatives to manual inspection by producing objective, repeatable assessments without exposing personnel to shaker-house hazards.

Vision Transformers (ViT)

Vision Transformers (ViT) have emerged as a powerful alternative to Convolutional Neural Networks (CNNs) for image-based tasks (Dosovitskiy et al. 2021). When processing images, ViT models segment an image into fixed-size patches, linearly embed them, and feed the resulting sequence into a transformer encoder, which captures global dependencies via self-attention (Vaswani et al. 2017; Khan et al. 2021). Unlike CNNs, which rely on local convolutional filters and strong inductive biases, ViTs have a weaker inductive bias but excel at learning long-range relationships when pre-trained on large datasets. Pre-trained ViTs have achieved state-of-the-art performance, sometimes surpassing CNNs in accuracy and efficiency during pre-training phases (Dosovitskiy et al. 2021; Touvron et al. 2020). Their capacity to model complex global patterns makes them particularly promising for analyzing the diverse visual features present on shale shakers.

Edge Computing in Industrial Applications

Edge computing brings computation and data storage closer to where data is generated in this case, the drilling rig equipment rather than relying on a centralized cloud (Alsheikh et al. 2021). By performing computations locally on edge devices, several advantages arise for industrial applications in remote or critical environments:

- Reduced Latency: On-site processing minimizes the delay between data acquisition and actionable insight, which is crucial for real-time monitoring and control.
- Bandwidth Conservation: Instead of transmitting raw high-resolution video to the cloud, which can be costly and unreliable in remote locations, edge computing sends only essential results or summaries.
- Enhanced Reliability and Autonomy: Operations can continue uninterrupted even if cloud connectivity is lost, supporting safety-critical systems and environments with intermittent communication.
- Improved Data Security and Privacy: Processing sensitive operational data locally reduces risks associated with transmitting it over potentially insecure networks.
- Lower Operational Costs: By minimizing data transmission and cloud-processing requirements, operators can achieve cost savings.

In the context of real-time shale shaker monitoring, deploying a ViT model on an edge device enables immediate analysis of camera feeds. This approach delivers timely alerts and insights to on-site personnel without dependence on high-bandwidth external networks, thereby addressing both safety and performance imperatives.

Objectives

The primary goal of this project was to develop and validate a machine-learning driven system for continuous, real-time monitoring of shale shaker conditions, with the ultimate aim of detecting early indicators of wellbore instability and optimizing drilling performance. Specifically, the system is designed to:

- Leverage high-resolution camera feeds and computer vision techniques to monitor shale shaker screens continuously.
- Classify shaker conditions into discrete categories (Empty, Fluid, Cuttings) with high accuracy.
- Detect deviations in shaker conditions with respect to drilling activities to signal emerging anomalies or wellbore instability.
- Integrate with edge-computing hardware to ensure on-site, instantaneous data processing, thereby eliminating delays associated with remote data transmission.
- Deliver value by:
  - Flagging early signs of abnormal shaker conditions linked to instability.
  - Enabling more effective hole cleaning through timely operational adjustments.
  - Reducing Non-Productive Time (NPT) by providing actionable insights before critical failures occur.

System Design and Architecture

Machine Learning System Architecture

The core of this solution is an in-house machine-learning pipeline that combines advanced image preprocessing with a Vision Transformer (ViT) classification model and correlates the predictions with rig activities. High-resolution cameras, mounted at strategic locations at the shale shaker, stream live video to an edge device. Each video frame undergoes the following steps:

- Region Masking: Irrelevant portions of the image (e.g., rig infrastructure beyond the shaker screen) are automatically masked so that computational resources focus exclusively on the zone where cuttings and fluid appear.
- Movement Detection: A lightweight motion-detection algorithm identifies regions of active particle flow across the screen, providing a secondary validation input that distinguishes between static artifacts and dynamically moving solids or fluid.
- ViT-Based Classification: Masked frames are subdivided into fixed-size patches, embedded, and processed by a fine-tuned Vision Transformer. The model assigns each frame to one of three classes:
  - Empty: No material present on the shaker screen.
  - Fluid: Drilling fluid visible without significant solids.
  - Cuttings: Drill cuttings visible on the shaker.

This three-class output is then timestamped and logged to a time-series database. The ViT model was selected because of its demonstrated ability to capture global image context, critical for distinguishing differences between fluid patterns and cuttings textures, while still running efficiently on edge-optimized hardware. The predictions are correlated with rig activities, and if anomalies are detected, alarms are raised to alert personnel.

Edge-Computing Deployment

To minimize latency and avoid the need for high-bandwidth uplinks, the entire vision pipeline (masking, motion detection, and ViT inference) executes on an edge device housed in a mobile trailer. Key design features include:

- On-Site Inference: A compact GPU/CPU accelerator board runs the ViT model locally, enabling classification at frame rates exceeding 10 fps without reliance on external servers.
- Autonomous Power Supply: The trailer can operate on solar power in remote locations or draw from rig power when available, ensuring uninterrupted monitoring even during outages.
- Low-Rate Metadata Streaming: Only classification results and summary statistics (e.g., particle counts, timestamps) are transmitted to the rig's control system or remote monitoring center, drastically reducing bandwidth consumption.

Methodology

Data Collection and Preprocessing

Data were captured during two trials on active drilling rigs.

- Trial 1: 15–17 July 2023, Rig A.
- Trial 2: 5–7 November 2023, Rig B.

Each trial employed high-resolution cameras fixed above a main shale shaker screen. Over 100 hours of continuous video were recorded, covering both day and night operations, varying cutting types, and fluctuating lighting conditions.

Preprocessing steps included.

- Frame Extraction & Labeling: Frames were extracted at 10 fps. Experts then manually labeled each frame into one of three classes Empty, Fluid, or Cuttings resulting in balanced datasets.
- Image Enhancement: Contrast adjustment and noise-reduction filters standardized image quality across different lighting conditions.
- Automated Masking: Using a static background reference, non-shaker regions were masked for some steps within the pipeline to eliminate visual clutter.
- Motion Detection: A background-subtraction algorithm identified moving pixels on the shaker screen, yielding a per-frame "particle-flow" count that was later correlated with model predictions to reduce false positives.

Model Development and Training

The classification model is based on a Vision Transformer backbone, chosen for its ability to learn global dependencies in visual data. The development process followed these steps:

Dataset Preparation.

- Training Set: 1,924 frames (Empty: 721; Fluid: 490; Cuttings: 713).
- Test Set: 483 frames (Empty: 181; Fluid: 123; Cuttings: 179).
- Classes were balanced and validated by two domain experts to eliminate ambiguous labels.

Transfer Learning & Fine-Tuning.

- A ViT pre-trained on a large-scale image corpus was fine-tuned using our drilling-rig dataset.
- Data augmentation (random rotations, flips, and brightness adjustments) increased robustness to lighting and orientation changes.
- Hyperparameters (learning rate, batch size, number of epochs) were optimized via grid search, using 20 percent of the training set as a validation split to detect overfitting early.

Results and Discussion

The developed system was rigorously evaluated based on its classification performance and its ability to provide actionable insights in real-world drilling scenarios.

Model Performance: Accuracy, Classification Report, and Confusion Matrix

The model's overall accuracy in correctly classifying images into the three categories (Empty, Fluid, Cuttings) on the test dataset was 100%. While a 100% accuracy score is the ideal outcome, it can sometimes warrant further scrutiny for potential methodological biases or overfitting, particularly with limited datasets. Therefore, additional testing was conducted, including hiding critical elements of the images to evaluate the model's areas of importance. These tests were completed successfully, and the model was accepted as 100% accurate on the available test dataset. No misclassifications were present during the evaluation of the test data.

Table 1—Classification report, summarizing the model's performance across the three classes. It includes key metrics such as precision, recall, and F1-score, all showing perfect values of 1.00, indicating the model's exceptional accuracy in correctly classifying all instances in the dataset. The support column reflects the number of samples for each class, and the overall accuracy is 100%, with the macro and weighted averages also achieving a score of 1.00 across the 483 samples.

Classification Report

	precision	recall	f1-score	support
Empty	1.00	1.00	1.00	181
Fluid	1.00	1.00	1.00	123
Cuttings	1.00	1.00	1.00	179
accuracy			1.00	483
macro avg	1.00	1.00	1.00	483
weighted avg	1.00	1.00	1.00	483

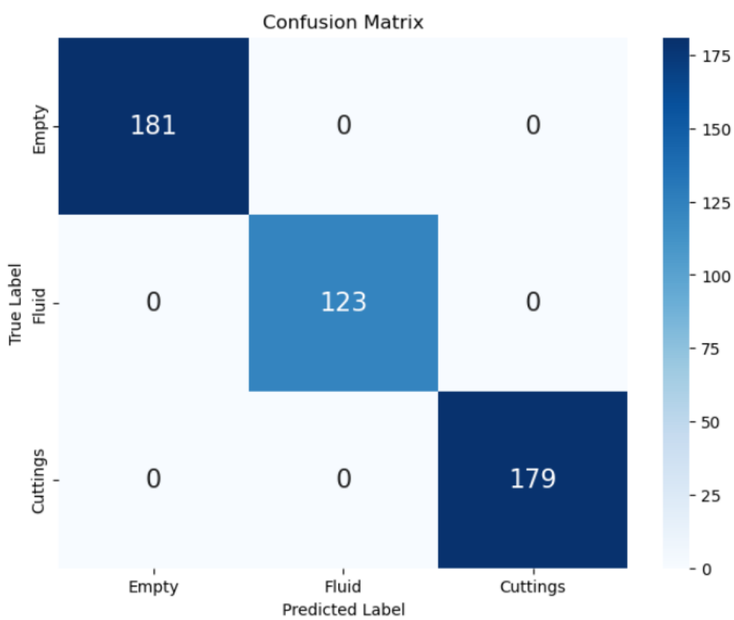

Figure 1—Confusion matrix, visually confirming the correct classification for all three classes.

Time-Series Analysis and Visualization of Operational Correlations

The model was evaluated using various samples of pre-recorded 10-minute videos, and corresponding time-series graphs of the predictions were produced. These graphs were then correlated with rig activities, illustrating the model's predictions over selected drilling operation periods. The analysis enabled trend identification, where the graphs showed clear correlations with drilling activities, such as the consistent presence of "Cuttings" during active drilling phases. The generated time-series data provides valuable operational insights into the shale shaker status while requiring minimal bandwidth, making it much easier to transmit to a remote operating center for effective monitoring of the drilling process.

The system's sensitivity to changes in the operational environment was highlighted by its ability to detect deviations from expected shaker conditions based on ongoing rig activity.

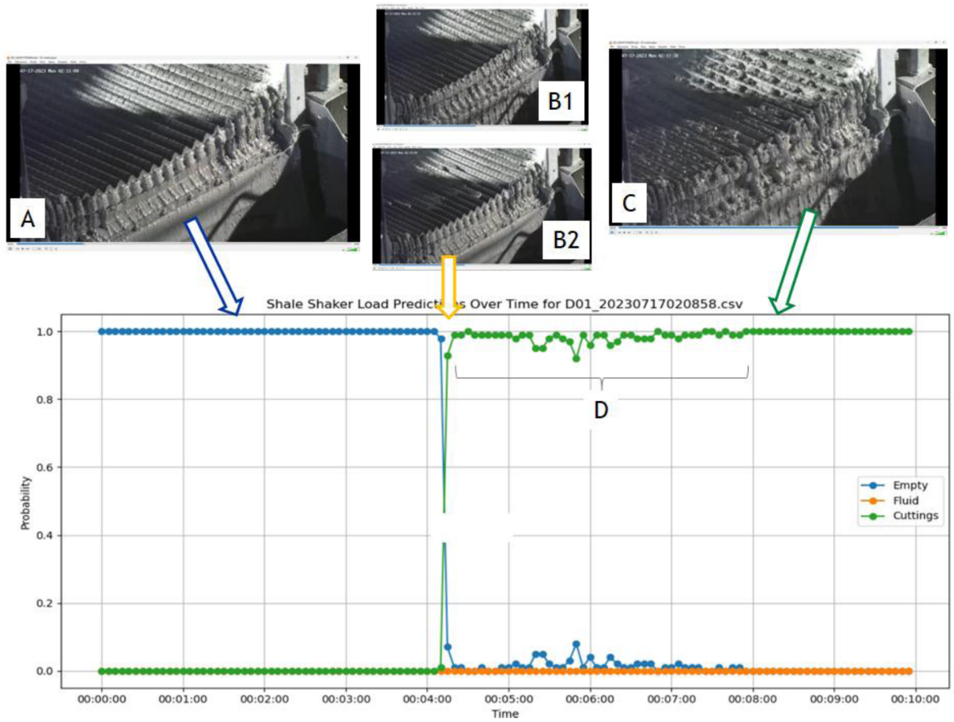

Figure 2—Time series with video snippets from a nightshift connection activity during drilling above. Label A indicates an empty shaker (no cuttings), B1 and B2 mark the transition zone, and C shows visible cuttings. Period D represents a lower volume of cuttings as drilling resumed after the connection.

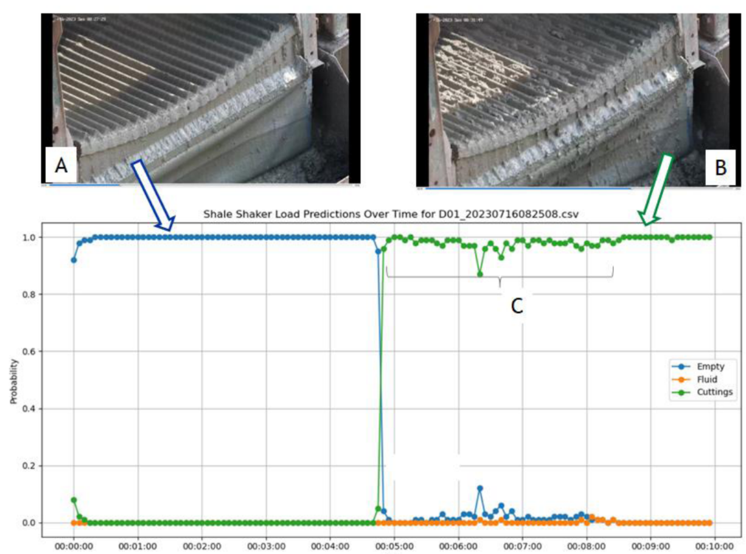

Figure 3—Time series with daylight connection. Label A shows an empty shaker (no cuttings), B indicates cuttings over the shakers with a transition zone, and C represents a zone with lower cuttings.

Anomaly Detection Capabilities

The system demonstrated its ability to detect anomalies, defined as deviations from expected shaker conditions based on ongoing rig activity.

Figure 4—Anomaly detection instance where, during standard drilling when a steady flow of cuttings was expected, a sudden drop in cuttings was observed. Investigation revealed this drop correlated with a technician washing the shaker screens, temporarily removing cuttings from view.

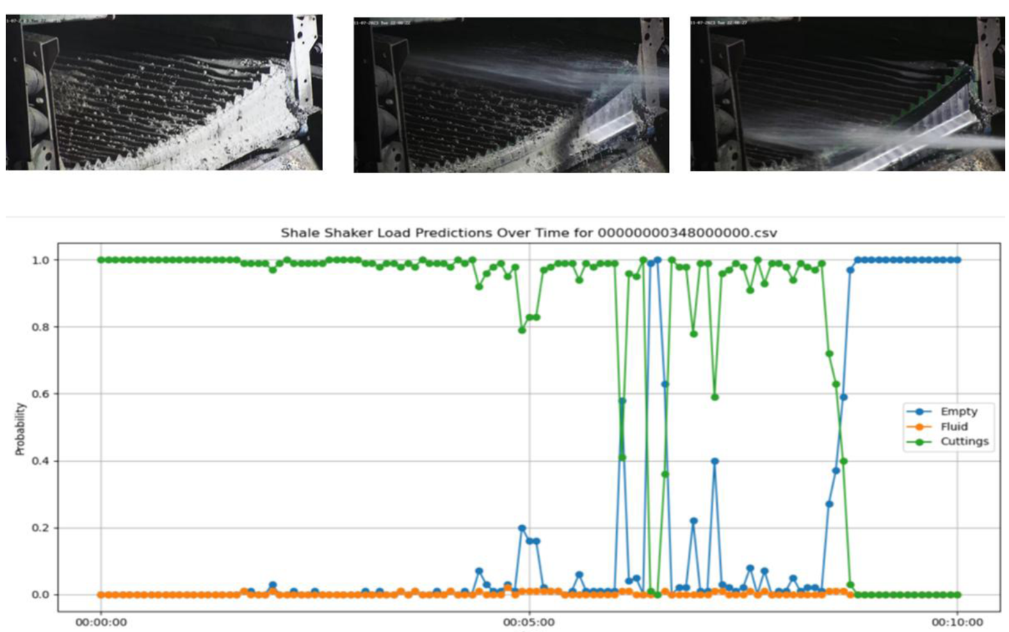

Figure 5—Similar anomaly where, during a technician's inspection and screen washing, the model detected no cuttings on the shaker screens when cuttings would normally be expected.

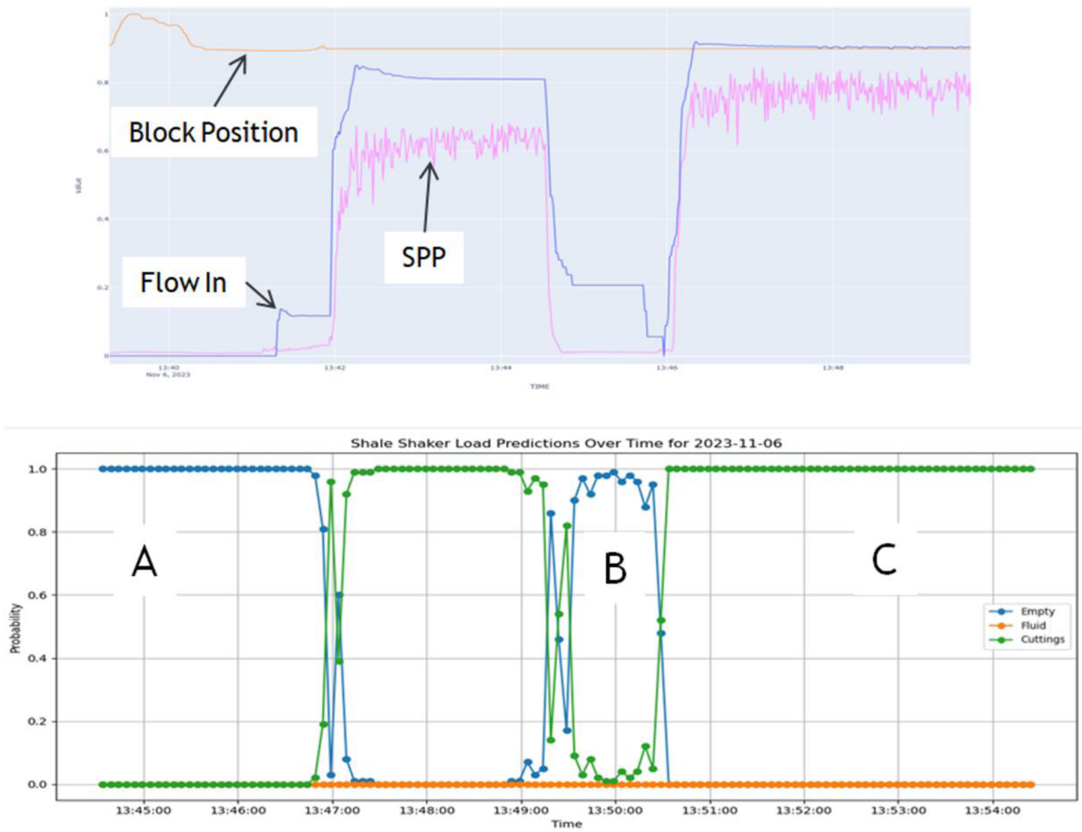

Figure 6—Correlation between rig activities recorded by sensors (top plot, with normalized Y-axis) and model predictions (bottom plot), aligned across the X-axis (time). In the top plot, Flow In, Standpipe Pressure, and Block Position are represented by blue, pink, and orange lines, respectively. The model's predictions align closely with the Flow In values. During a connection (label A), when fluid circulation stopped, no cuttings were detected, as expected. At label B, the model again detected no cuttings, correlating with a post-connection survey. After the survey, drilling resumed, and the model accurately detected continuous cuttings at label C.

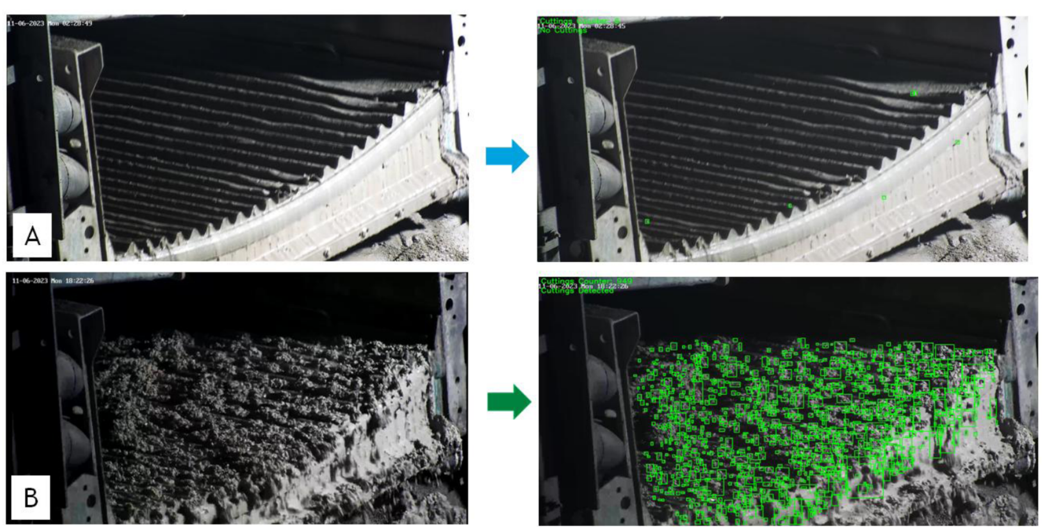

Figure 7—Incorporated movement detection in the zone of interest. In Label A, no cuttings are present, and the model detects 6 moving particles across the shale shaker. In Label B, with cuttings present, the model detects 949 moving particles, displayed in the top left corner of the bottom right frame. The applied thresholding for moving objects helps differentiate true material flow from static screen conditions or minor fluid sheen, thereby enhancing the classifier's robustness by providing a dynamic context to the visual classification and reducing potential false positives from static artifacts.

Enhancing Anomaly Interpretation through Advanced Visual Analysis

While the current system effectively detects deviations from expected shaker conditions, a key area for future enhancement lies in automatically distinguishing genuine anomalies from planned human interventions, such as screen cleaning or maintenance. The system's current response to technician activities, logging changes in shaker state like "Empty" or unusual patterns validate its sensitivity to any visual alteration. However, by integrating more advanced computer vision techniques, the system could provide richer contextual understanding.

Future iterations could incorporate object detection models like YOLO (You Only Look Once) to identify the presence of personnel near or on the shakers. This would be a first step in flagging that an observed change might be due to human activity. Taking this further, adding a vision-to-text model within the data processing pipeline, especially during anomaly detection flags, could help automatically describe the scene or recognize when anomalies are desired maintenance actions rather than unintended operational deviations. For instance, a vision-to-text model might generate descriptions like "technician cleaning shaker screen" or "personnel replacing screen mesh."

While more computationally intensive and perhaps less suited for immediate edge deployment without further optimization, Visual Large Language Models (VLLMs) could eventually offer even more sophisticated scene interpretation, potentially understanding complex actions and their implications directly from the visual feed. The immediate goal, however, would be to leverage more lightweight person detection and activity recognition (via vision-to-text or specialized action recognition models) to reduce false alarms for process issues when routine maintenance is underway. This would allow operations teams to focus on true unexpected anomalies, further refining the system's utility in proactive drilling management.

Limitations of the Current Study

The current results are promising, however this study has several limitations that provide direction for future work:

- Limited Dataset Scope: The model was trained and validated using data from only two wells.
- Robustness to Extreme Environmental Conditions: The impact of extreme camera fouling (e.g., heavy mud splatter, salt spray build-up) or severe vibration beyond what was encountered requires more extensive testing. While enclosures offer protection, their long-term efficacy and maintenance requirements in very harsh conditions need to be assessed.
- Fixed Classification Categories and Granularity: The current three-class system (Empty, Fluid, Cuttings) is effective for initial status determination but could be significantly expanded to provide more nuanced insights. Future work should focus on increasing the granularity of classifications. This includes:
    - Differentiating Cuttings Types: Distinguishing between normal drilled cuttings and cavings indicative of instability would be a major step towards direct wellbore health monitoring.
    - Identifying Specific Issues: Adding classes for conditions like screen blinding, screen damage, or fluid overload as distinct categories would enhance diagnostic capabilities.
    - Quantifying Cuttings Volume Levels: Instead of a single "Cuttings" class, a multi-level classification for cuttings volume could be implemented. For example, the next phase of testing could introduce five distinct levels: No Cuttings (Empty), Trace Cuttings, Low Volume Cuttings, Normal Volume Cuttings, and High Volume (Overload) Cuttings. This would allow the system to track not just the presence, but also the relative flow rate of cuttings over the shakers, providing early warnings of insufficient hole cleaning (decreasing volume) or potential wellbore events leading to excessive cuttings (increasing volume or overload). This finer-grained volume

assessment would offer a more dynamic and quantitative view of hole cleaning efficiency and downhole conditions.

- Integration with Other Sensors: While powerful as a standalone vision system, its value could be further amplified by direct integration and correlation with other real-time drilling data sources (e.g., MWD/LWD data, surface mud logging gas, pit volumes) within a unified analytical platform.

Novelty and Additive Value

This paper introduces a significant advancement in the monitoring of shale shakers and, by extension, in the early detection of conditions indicative of potential wellbore instability. The novelty of the presented system lies in its holistic approach, which transforms a traditionally manual and intermittent process into a fully automated, continuous, and real-time monitoring solution.

The core additive value stems from this fundamental shift. Traditional shale shaker monitoring relies on periodic visual checks by rig personnel, which are inherently subjective, prone to human inconsistency, and offer only snapshots in time. This often leads to delayed recognition of critical changes in cuttings return or shaker performance. In contrast, our system provides an uninterrupted, objective data stream, enabling immediate awareness of the shaker status. This continuous vigilance is crucial for capturing transient events or subtle trends that might be missed by manual inspections but could be early indicators of developing downhole issues.

The uniqueness of this work is further underscored by the synergistic integration of three key technologies for this specific application:

- High-Resolution Imaging: Purpose-selected and strategically positioned cameras capture detailed visual information of the shaker screens, providing a rich data source far exceeding simple sensor readings.
- Vision Transformer (ViT) Architecture: The application of a state-of-the-art ViT model for image classification allows the system to distinguish complex visual patterns and accurately categorize shaker conditions ("Empty," "Fluid," "Cuttings") with a high degree of precision, as demonstrated by the performance results and potential for future enhancements.
- Edge Computing Deployment: Processing the imaging data and ML model inference directly on-site using an edge device is a critical enabler. This ensures real-time analysis without reliance on potentially unreliable or high-cost rig-to-shore data links, making the system robust, responsive, and self-sustained.

This combination of continuous visual data acquisition, advanced AI-driven interpretation, and localized real-time processing sets a new benchmark for shale shaker monitoring. Previously, achieving such a level of detailed, autonomous, and immediate insight into shaker operations was not practically feasible. By automating this crucial monitoring task, the system not only enhances the potential for early detection of wellbore instability and optimization of hole cleaning but also frees up rig personnel for other critical duties. It moves beyond simple alarm systems to provide a continuous, classified understanding of a key component of the drilling fluid circulation system, paving the way for more data-driven and proactive drilling operations. The potential to reduce operational downtime, improve safety, and enhance overall drilling efficiency establishes this system as a valuable and innovative contribution to the industry.

Conclusions and Future Work

This paper successfully detailed the development, deployment, and validation of a novel, real-time shale shaker monitoring system leveraging Vision Transformer (ViT) models and edge computing. Key achievements include the engineering of a complete deployable system, high classification accuracy (100% on the test set for "Empty," "Fluid," "Cuttings" states), and demonstrated real-time processing capabilities independent of external network constraints. The system's output showed strong correlation with rig activities and sensitivity to operational changes.

The significance of this work lies in transforming manual, intermittent shaker monitoring into an automated, continuous, and objective process, setting a new benchmark for operational awareness and early detection of conditions relevant to wellbore instability. This offers tangible benefits such as enhanced safety, potential NPT reduction, and optimized hole cleaning.

Future work

- Dataset Expansion: Broadening data acquisition across diverse wells, geologies, mud types, and shaker designs to enhance model robustness and generalization.
- Granular Classification: Refining the model to differentiate cuttings types (e.g., normal vs. cavings), identify specific faults, quantify cuttings volume through multiple levels, and predict performance trends.
- Contextual Anomaly Recognition: Incorporating vision-to-text or object detection models to better distinguish planned maintenance from true operational anomalies, thereby minimizing false positives.

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